

D E C K 1 4 · W E E K 4 · A F T E R N O O N 2

Fixed Income II

Pricing, Yields & the Term Structure

From a single discount rate to a full spot / par / forward curve

CFA Level I · Vol 6 Learning Modules 6–9

Roadmap — Fixed Income II: Pricing, Yields & Term Structure

4 sections · ~2 hours · Built bottom-up from CFA readings

01

Bond pricing mechanics

PV of cash flows · discount/par/premium · flat vs full price · accrued interest · matrix pricing

02

Yield spread measures · fixed-rate

current yield · street convention · yield-to-worst · G-spread · I-spread · Z-spread · OAS

03

Money-market and floater yields

discount rate vs add-on rate · bond-equivalent yield · FRN pricing · discount margin

04

Term structure: spot / par / forward

spot curve · par rates · implied forwards · bootstrapping · curve-shape relationships

Learning objectives

By the end of this deck you should be able to:

1. Price a fixed-coupon bond on and between coupon dates, split full price into flat price and accrued interest, and classify it as discount, par or premium.
2. State the three conditions under which an investor actually earns the quoted yield-to-maturity, and estimate the price of an illiquid bond by matrix pricing.
3. Compute current yield, street-convention YTM, yield-to-call and yield-to-worst on a callable bond, and reconcile yields quoted at different periodicities.
4. Decompose a yield into benchmark plus G-spread, I-spread or Z-spread, and explain how an OAS adjusts the Z-spread for an embedded option.
5. Convert money-market discount rates to bond-equivalent yields, solve an FRN for its discount margin, and move between spot, par and forward curves.

SECTION 01

Bond pricing mechanics

A bond's price is the present value of its promised cash flows discounted at the market discount rate — and moves inversely with that rate

The market discount rate r is the return investors demand for the bond's credit, liquidity and maturity risks — also called the required yield.

$$PV = PMT_1 / (1 + r)^1 + PMT_2 / (1 + r)^2 + \dots + (PMT_n + FV_n) / (1 + r)^n$$

where: PMT_t = coupon per period; FV_n = face value at maturity; r = market discount rate PER PERIOD; n = number of periods to maturity.

RATE $\uparrow$ $\rightarrow$ PRICE $\downarrow$

Discount factors $(1+r)^t$ grow with r ; a higher rate shrinks every future cash flow by more. The cash flows never move — only the denominators.

CONVEX, NOT LINEAR

The price–yield line is curved. An 80-bp fall raises the price by MORE than an 80-bp rise cuts it. Positive convexity — Deck 15.

MATURITY AND COUPON

The longer the maturity and the lower the coupon, the larger the % price change for a given Δr . That is the duration intuition, arriving early.

Source: CFA Institute, Curriculum Vol 6, Learning Module 6, §2 "Bond Pricing and the Time Value of Money", Equations 1–2, pp. 129–132.

Coupon < YTM → discount; coupon = YTM → par; coupon > YTM → premium — three regimes, and the price gap is the PV of the coupon shortfall or excess

BRWA five-year bond, 3.2% p.a. paid semiannually (1.6% per period), priced at three different per-period market discount rates.

Periodic YTM	Coupon vs rate	Price	Regime	Where the gap comes from
1.2%	Coupon > rate	103.75	Premium	PV of ten +0.4 excess coupons = +3.75
1.6%	Coupon = rate	100.00	Par	No excess, no shortfall – price = 100
2.0%	Coupon < rate	96.41	Discount	PV of ten -0.4 shortfalls = -3.59

PULL-TO-PAR

Hold the YTM constant and a discount bond drifts UP toward 100, a premium bond drifts DOWN toward 100. Total return = coupon income + this predictable capital gain or loss.

YTM — THE THREE CONDITIONS

You earn the quoted YTM only if (1) you hold to maturity, (2) the issuer pays every coupon and the principal in full and on time, and (3) you reinvest every coupon AT the YTM.

ANNUALIZED STATEMENT

Same fact stated per annum: BRWA is at a premium because its 3.2% stated coupon beats the 2.4% stated market discount rate. Periodic rate $\times m$ = stated annual rate.

Source: CFA Institute, Curriculum Vol 6, LM 6, §2, Example 1 (BRWA bond), Exhibits 1–3, pp. 130–132.

Worked — a five-year 3.25% semiannual bond at a 4.0% market discount rate prices at 96.632, and the 3.368 discount is exactly the PV of the deficient coupons

Coupon 3.25% < discount rate 4.0% → the bond must trade below par. Convert to PERIODS before you touch the calculator.

$$P = PMT \cdot [1 - (1+r)^{-n}] / r + FV / (1 + r)^n$$

where: Annuity form for level coupons plus a single PV of face. Identical to summing the cash flows one by one — just faster.

CFA LM 6 — 5-YEAR 3.25% SEMIANNUAL CORPORATE BOND • 30/360

Face 100, coupon 3.25% p.a. paid semiannually, market discount rate 4.0% p.a. Compute the price.

1. Convert first: $n = 5 \times 2 = 10$ periods · $PMT = 3.25 / 2 = 1.625$ · $r = 4.0\% / 2 = 2.0\%$ per period
2. PV of coupons: $1.625 \times [1 - (1.02)^{-10}] / 0.02 = 1.625 \times 8.98259 = 14.597$
3. PV of face: $100 / (1.02)^{10} = 100 / 1.21899 = 82.035$
4. Price = $14.597 + 82.035 = 96.632$ per 100 of par value
5. Cross-check: deficient coupon = $(4.00 - 3.25)/2 = 0.375$ per period → $0.375 \times 8.98259 = 3.368 = 100 - 96.632$ ✓

P = 96.632 per 100 face · the 3.368 discount IS the PV of ten 0.375 coupon shortfalls

Source: CFA Institute, Curriculum Vol 6, LM 6, Learning Module Self-Assessment Q1, pp. 126–127; cross-check against §2 Exhibit 1, p. 131.

Between coupon dates full price = flat price + accrued interest — dealers quote the flat price but the buyer settles on the full price

Quote full prices and the screen climbs every day, then gaps down on each coupon date — a sawtooth from the calendar.

$$\begin{aligned} \text{PV_Full} &= \text{PV_Flat} + \text{AI} & \text{AI} &= (t / T) \times \text{PMT} \\ \text{PV_Full} &= \text{PV_last-coupon-date} \times (1 + r)^{(t/T)} \end{aligned}$$

where: t = days elapsed since the last coupon; T = days in the full coupon period; PMT = coupon per period. Day count is Act/Act or 30/360.

FLAT (CLEAN)

What dealers show on the screen, and what every price statistic and spread is computed from. Strips out the mechanical accrual, so a price move means a yield move.

FULL (DIRTY)

The cash the buyer actually wires at settlement. Includes the slice of the next coupon earned by the seller. Depends on the calendar only — never on the yield.

DAY COUNT SETS t / T

Act/Act uses real days and a real year (governments). 30/360 assumes 30-day months and a 360-day year (corporates). CFA Example 2: the same accrual is 43/184 on one and 42/180 on the other.

Source: CFA Institute, Curriculum Vol 6, LM 6, §2 "Flat Price, Accrued Interest, and the Full Price", Equations 3–6 and Exhibit 5, pp. 134–136.

Worked — the Romania 4.625% EUR bond settling 15 Dec 2031 at a 3.50% YTM: full price 117.635, accrued 3.235, flat price 114.400

Annual coupons on 3 April. Settlement on 15 December sits 256 days into a 366-day coupon period — 2032 is a leap year.

ROMANIA 4.625% EUR • ANNUAL • ACT/ACT • MATURES 3 APRIL 2049

Settlement 15 Dec 2031, YTM on that date 3.50%. Eighteen full annual coupons remain (April 2032 through April 2049).

1. Price it on the LAST coupon date, 3 Apr 2031: $PV = -PV(0.035, 18, 4.625, 100) = 114.838$
2. Count days: 3 Apr → 15 Dec 2031 = 256 days; full period 3 Apr 2031 → 3 Apr 2032 = 366 days
3. Roll forward: $PV_Full = 114.838 \times (1.035)^{(256/366)} = 114.838 \times 1.02435 = 117.635$
4. Accrued interest: $AI = (256/366) \times 4.625 = 0.69945 \times 4.625 = 3.235$
5. Flat (quoted) price: $PV_Flat = 117.635 - 3.235 = 114.400$

PV_Full 117.635 • AI 3.235 • PV_Flat 114.400 – the screen shows 114.400

WHY THE QUOTED PRICE IS THE FLAT PRICE

In a world with an unchanged yield, the flat price barely moves across the coupon date while the full price falls by the whole coupon. Quoting flat means a change in the number means something.

Matrix pricing estimates the yield of an illiquid or unissued bond from a grid of actively traded comparables — a four-step process

Unlike listed equities, most bonds are not actively traded — and a bond that has not been issued yet has no price at all. Comparables fill the gap.

YTM_{target} ≈ linear interpolation of comparable YTM_s by time-to-maturity

- 1 IDENTIFY COMPARABLES**
Similar time-to-maturity, coupon rate and credit quality. Recent prints only.
- 2 BACK OUT EACH YTM, THEN AVERAGE**
Solve each comparable price for its yield, then average the yields within each maturity bucket.
- 3 INTERPOLATE TO THE TARGET TENOR**
Linear interpolation between the bracketing buckets gives the target bond's estimated yield.
- 4 PRICE THE TARGET BOND**
Discount the target bond's own coupons and principal at that estimated yield.

Source: CFA Institute, Curriculum Vol 6, LM 6, §4 "Matrix Pricing", Exhibits 8–11, pp. 146–149.

Worked — an illiquid four-year 4.50% bond prices at 104.99 off a three-year 5.50% at 107.500 and a five-year 4.50% at 104.750

Two traded comparables of similar credit quality bracket the target maturity — a clean two-point ladder to interpolate across.

MATRIX-PRICING AN ILLIQUID 4-YEAR 4.50% CORPORATE BOND

Comparable A: 3-year, 5.50% annual, priced 107.500. Comparable B: 5-year, 4.50% annual, priced 104.750. All annual compounding.

1. 3-year yield: $107.500 = 5.50/(1+r) + 5.50/(1+r)^2 + 105.50/(1+r)^3 \rightarrow r_3 = 2.856\%$
2. 5-year yield: $104.750 = 4.50/(1+r) + \dots + 104.50/(1+r)^5 \rightarrow r_5 = 3.449\%$
3. Interpolate to 4 years: $r_4 = (0.02856 + 0.03449) / 2 = 0.031525 = 3.153\%$
4. Price the target at 3.1525%: $4.50/(1.031525) + 4.50/(1.031525)^2 + 4.50/(1.031525)^3 + 104.50/(1.031525)^4$
5. Sum the four PVs: $4.362 + 4.229 + 4.100 + 92.299 = 104.991$

Estimated price = 104.991 per 100 of par · estimated yield 3.153% · a premium

INTERPOLATE YIELDS, NEVER PRICES

The two comparables carry different coupons, so their prices are not comparable — only their yields are. Averaging the two prices gives 106.125, which is exactly one of the exam distractors.

SECTION 02

Yield spread measures for fixed-rate bonds

Current yield, day counts, street convention, true yield and government-equivalent yield are all one theme — what we discount and over what calendar

Most of these differ by a basis point or two — but the difference is always material the moment you subtract one bond's yield from another's.

Measure / convention	What it means	Where used
Current yield	Annual coupon / flat price — ignores time value	Quick proxy
Act/Act day count	Actual days / actual days in the period	Governments
30/360 day count	30-day months, 360-day year	Corporates
Street convention	Cash flows assumed on scheduled dates	Dealer quotes
True yield	Actual pay dates — never above street yield	Rarely quoted
Government-equivalent	30/360 corporate YTM restated on Act/Act	G-spread input
Simple yield	Coupon + straight-line amortisation / flat price	Japan (JGBs)

PERIODICITY CONVERSION — NEVER SKIP IT

$(1 + \text{APR}_m/m)^m = (1 + \text{APR}_n/n)^n$. CFA Example 5: the Antelas AG quarterly-pay bond yields 4.548% at periodicity 4, which is 4.574% at periodicity 2. Align periodicities before you difference two yields.

Source: CFA Institute, Curriculum Vol 6, LM 7, §3 "Other Yield Measures and Conventions", Exhibit 3, Example 4 and Example 5, pp. 163–167.

Yield-to-worst on a callable bond is the lowest of the yield-to-maturity and every yield-to-call – the conservative yield a buyer should assume

VIVU 6.5% seven-year callable notes priced at 106.50, semiannual coupons of 3.25. Call protection for three years, then a step-down call schedule.

$$\begin{aligned} PV &= \sum PMT / (1+r)^t + (PMT + \text{Call price}) / (1+r)^N \\ YTW &= \min \{ YTC_1, YTC_2, \dots, YTM \} \end{aligned}$$

where: The issuer calls only when it suits the issuer — which is when it hurts the investor. So assume the worst outcome, not the best.

Scenario	Redeemed	Periods	Redemption	Semi r	Annualized
Yield-to-first call	End yr 3	6	103.25	2.5748%	5.15%
Yield-to-second call	End yr 4	8	102.50	2.6237%	5.25%
Yield-to-third call	End yr 5	10	101.75	2.6530%	5.31%
Yield-to-fourth call	End yr 6	12	101.00	2.6811%	5.36%
Yield-to-maturity	End yr 7	14	100.00	2.6868%	5.37%
Yield-to-worst	—	—	—	—	5.15%

Source: CFA Institute, Curriculum Vol 6, LM 7, §3 Example 6 (VIVU seven-year callable notes) and Exhibit 4, pp. 167–169.

Every YTM decomposes into a benchmark rate plus an issuer-specific spread — macro sits in the benchmark, micro sits in the spread

Benchmark = expected real rate + expected inflation. Spread = credit + liquidity + taxation. Analysts track the spread against its own history, not the absolute yield.

$$\begin{aligned}\text{YTM}_{\text{bond}} &= \text{Benchmark yield} + \text{Yield spread} \\ \text{G-spread} &= \text{YTM}_{\text{bond}} - \text{government YTM}\end{aligned}$$

where: Benchmark = the on-the-run sovereign of matching maturity, interpolated when no matching bond exists. Spreads are quoted in basis points.

G - S P R E A D

Over an actual or interpolated government bond of the same maturity. The default in the US, UK and Japan. One number minus one number.

I - S P R E A D

Over the matched-tenor interest-rate swap rate. The convention for euro-denominated corporates, quoted as "mid-swaps plus X".

Z - S P R E A D

A constant add-on to EVERY benchmark spot rate such that the PV of the cash flows equals the price. Uses the whole curve, not one point.

Source: CFA Institute, Curriculum Vol 6, LM 7, §4 "Yield Spreads over Benchmark Rates", Exhibit 5 and Exhibit 8, pp. 171–177.

Worked — the Russian Federation 5.25% 30-year bond at 123.5 with 24 years left has a G-spread of 166 bps over an interpolated 24-year Treasury

Issued at 99.625 at 200 bps over the 30-year Treasury. Six years on, no 24-year Treasury exists — so we build one by interpolation.

RUSSIAN FEDERATION 5.25% USD · SEMIANNUAL · 24 YEARS LEFT

On 15 Mar 2026 the bond trades at 123.5. The 20-year US Treasury yields 2.00% and the 30-year yields 2.25%. Find the current G-spread.

1. Current YTM: solve $123.5 = \sum 2.625/(1+r)^t + 100/(1+r)^{48}$ over 48 semiannual periods $\rightarrow$ YTM = 3.756%
2. Interpolation weights: $w_{20} = (30 - 24)/(30 - 20) = 60\%$, $w_{30} = 40\%$ [check: $0.6 \times 20 + 0.4 \times 30 = 24$]
3. Synthetic 24-year benchmark: $r_{24} = (60\% \times 2.00\%) + (40\% \times 2.25\%) = 2.10\%$
4. G-spread = $3.756\% - 2.10\% = 1.656\% = 166$ bps
5. 166 bps today against 200 bps at issuance $\rightarrow$ the market charges this issuer less than it did

G-spread = 166 bps · 34 bps tighter than the 200 bp issuance spread

WHAT A G-SPREAD CANNOT SEE

It uses one point on the government curve and one yield on the bond, so it says nothing about the shape of the curve in between. On a steep curve, that blind spot is exactly what the Z-spread closes.

Source: CFA Institute, Curriculum Vol 6, LM 7, §4 Example 7 "Calculating G-Spread without a Comparable Benchmark Bond", pp. 174–175.

The Z-spread is the constant add-on to every benchmark spot rate that makes the PV of the bond's cash flows equal its price — and OAS strips the option out of it

G-spread differences two single yields. Z-spread never averages the curve — it adds the same number to every point on it, then solves.

$$\begin{aligned}\text{Price} &= \sum_t \text{PMT} / (1 + z_t + Z)^t + (\text{PMT} + \text{FV}) / (1 + z_n + Z)^n \\ \text{OAS} &= \text{Z-spread} - \text{option value (bps)}\end{aligned}$$

where: z_t = benchmark spot rate for period t ; Z = the Z-spread, identical in every period, solved numerically. Also called the static or zero-volatility spread.

G USES ONE RATE

A YTM is one PV-weighted average of the curve, and the weights depend on the coupon. So G compares two slightly different averages. Z adds a constant to every spot and averages nothing.

ZERO-VOLATILITY

Z assumes one deterministic path for rates — no volatility, so no option value. That is exactly why a bond with an embedded option needs a further adjustment.

OAS SIGN RULE

Callable: you SOLD the option, part of the Z-spread pays you for it → $\text{OAS} < Z$.
Putable: you OWN the option → $\text{OAS} > Z$. OAS is the like-for-like credit spread.

Source: CFA Institute, Curriculum Vol 6, LM 7, §4 "Yield Spreads over the Benchmark Yield Curve", Equations 4–5 and Exhibit 8, pp. 176–177.

Worked — a three-year 5.5% corporate bond at a 378 bp Z-spread over government spots of 2.98 / 3.48 / 4.00% prices at 94.20

Each cash flow is discounted at its OWN government spot rate plus the SAME Z. On the desk you run this backwards: quote the price, solve for Z.

3 - YEAR 5.5 % CORPORATE BOND • Z - SPREAD 378 BPS

Government spot curve (effective annual): $z_1 = 2.98\%$, $z_2 = 3.48\%$, $z_3 = 4.00\%$. Cash flows 5.5, 5.5, 105.5. Find the price.

1. Build each discount rate: $z_1 + Z = 6.76\%$ · $z_2 + Z = 7.26\%$ · $z_3 + Z = 7.78\%$ (same 3.78% added to each)
2. Year 1: $5.5 / (1.0676)^1 = 5.152$
3. Year 2: $5.5 / (1.0726)^2 = 5.5 / 1.15047 = 4.781$
4. Year 3: $105.5 / (1.0778)^3 = 105.5 / 1.25203 = 84.263$
5. Price = $5.152 + 4.781 + 84.263 = 94.20$ — a deep discount, since 5.5% coupon vs 6.8–7.8% discount rates

Price = 94.20 per 100 of par • run it in reverse and 94.20 returns Z = 378 bps

ON A SCREEN IT RUNS THE OTHER WAY

The price is the observable and Z is the unknown: Solver or Goal Seek, objective = the quoted price, change cell = Z. That is why credit is compared in Z-spread — it is the only measure comparable across coupons and maturities.

Source: CFA Institute, Curriculum Vol 6, LM 7, §4 Question Set Q2 (Mexican corporate bond), Equation 4, p. 179.

SECTION 03

Money-market and floater yields

Money-market rates are annualized but NOT compounded — and come in two mutually inconsistent flavours: discount rates on FV and add-on rates on PV

Bond yields are annualized AND compounded. Money-market yields are simple interest — and two instruments with identical cash flows will quote different numbers.

$$\begin{aligned} \text{PV_DR} &= \text{FV} \times [1 - (\text{Days/Year}) \times \text{DR}] \\ \text{PV_AOR} &= \text{FV} / [1 + (\text{Days/Year}) \times \text{AOR}] \end{aligned}$$

where: DR is quoted on the face value paid at maturity; AOR is quoted on the price at issuance. The year is 360 or 365 depending on instrument and jurisdiction.

Convention	Formula	Typical instruments
Discount rate (DR)	$\text{DR} = (\text{Year/Days}) \times (\text{FV} - \text{PV}) / \text{FV}$	T-bills • CP • BAs
Add-on rate (AOR)	$\text{AOR} = (\text{Year/Days}) \times (\text{FV} - \text{PV}) / \text{PV}$	CDs • repos • MRRs
Bond-equivalent yield (BEY)	AOR stated on a 365-day year	Common basis

FOUR WAYS TO SAY THE SAME THING

Quote basis (DR or AOR) × year basis (360 or 365) = four dialects for identical cash flows. In CFA Example 4 the 180-day instrument with the LOWEST quoted rate, 4.33%, turns out to have the HIGHEST bond-equivalent yield at 4.487%.

¹ A discount rate divides the interest earned by the maturity value, not by the amount invested — so by design it understates the true return.

² Convert everything to a 365-day add-on rate (the bond-equivalent yield) before comparing any two money-market instruments.

Source: CFA Institute, Curriculum Vol 6, LM 8, §3 "Yield Measures for Money Market Instruments", Equations 2–5, Exhibits 2–3 and Example 4, pp. 194–199.

Worked — a 90-day CP quoted at a 0.120% discount rate on a 360-day year is a 0.122% bond-equivalent yield, which beats a CD quoting 0.120% add-on

Same credit, same 90-day tenor, two quoting conventions. Until they are translated, the two numbers cannot be compared at all.

BRWA 90-DAY COMMERCIAL PAPER vs CFP BANK 90-DAY CD

CP: 0.120% discount rate, 360-day year. CD: 0.120% add-on rate, 365-day year. Which offers the higher return?

1. Price the CP from its discount rate: $PV = 100 \times [1 - (90/360) \times 0.00120] = 99.970$
2. Restate on a 365-day add-on basis (Equation 5): $AOR = (365/90) \times (100 - 99.970) / 99.970$
3. $AOR = 4.05556 \times 0.00030009 = 0.00122$ — and that add-on rate IS the bond-equivalent yield
4. The CD is already an add-on rate on a 365-day year, so its 0.120% needs no conversion
5. Both adjustments — FV→PV in the denominator, 360→365 in the year — push the same way, so a DR always converts UP

CP BEY 0.122% vs CD 0.120% → the CP wins by ~0.2 bps once conventions match

THE QUOTED NUMBER IS NOT THE RETURN

Two quotes that both read "0.120%" are not the same return. Until both are expressed as 365-day add-on rates, the comparison is meaningless — and the ranking can flip.

¹ The printed setup line reads 0.100%, but the curriculum's own solution and its 0.2 bps conclusion both use DR = 0.120%. We follow the calculation.

Source: CFA Institute, Curriculum Vol 6, LM 8, §3 Example 3 and Exhibit 3, pp. 197–198.

A floating-rate note resets its coupon each period to MRR + quoted margin — so it prices at par on a reset date only when the required margin equals the quoted margin

The coupon snaps back to market every reset, so interest-rate risk is small. What is left is CREDIT risk — and that is what the discount margin measures.

$$PV = \sum \left[\frac{(MRR + QM) \times FV / m}{1 + (MRR + DM)/m} \right]^t + \frac{FV}{1 + (MRR + DM)/m}^N$$

where: QM = quoted margin, fixed at issuance, sits in the NUMERATOR. DM = discount margin, set by the market today, sits in the DENOMINATOR. Par when they are equal.

DM > QM → DISCOUNT

Credit has deteriorated. The issuer now pays a "deficient" margin versus what the market demands, so the note falls below par by the PV of that shortfall.

DM = QM → PAR

Nothing has changed since issuance. The note prices exactly at 100 on every reset date, and is pulled back to par between them.

DM < QM → PREMIUM

Credit has improved. The issuer is now "overpaying" versus the market, so the note trades above par by the PV of that excess margin.

Source: CFA Institute, Curriculum Vol 6, LM 8, §2 "Yield and Yield Spread Measures for Floating-Rate Notes", Equation 1 and Exhibit 1, pp. 188–190.

Worked — the Antelas AG four-year FRN paying 3-month MRR of -0.55% plus 250 bps, priced at 97, implies a discount margin of 329 bps

The note sits 3 points below par, so the market must be demanding more than the 250 bp locked in at issuance. Solve the pricing model for the margin that produces 97.

ANTELAS AG FRN • EUR • QUARTERLY • 4 YEARS • 30/360

QM = 250 bps. Three-month MRR held constant at -0.55% . Price 97 per 100 of par. $m = 4$, $N = 16$ periods, $FV = 100$.

1. Periodic coupon: $PMT = (MRR + QM) \times FV / m = (-0.0055 + 0.0250) \times 100 / 4 = 0.4875$
2. Solve for the periodic discount rate: $PV = -97$, $FV = 100$, $PMT = 0.4875$, $N = 16 \rightarrow r = 0.686122\%$
3. Unwind r into the margin: $r = (MRR + DM) / m \rightarrow 0.00686122 = (-0.0055 + DM) / 4$
4. $DM = 4 \times 0.00686122 + 0.0055 = 0.03294 = 329$ bps
5. Check: the 79 bp shortfall is 0.1975 a quarter; $\times$ the 16-period annuity factor 15.10 = 2.98 $\approx$ the 3-point discount ✓

DM = 329 bps vs QM 250 bps $\rightarrow$ the 79 bp shortfall IS the 3-point discount

WATCH THE SIGNS AND THE PERIODICITY

MRR is negative here, so $-(-0.0055)$ adds back. And r is a PERIODIC rate — multiply by $m = 4$ before subtracting MRR. Dropping either step is the standard way this calculation goes wrong.

Source: CFA Institute, Curriculum Vol 6, LM 8, §2 Example 1 "Calculating the Discount Margin for a Floating-Rate Note", pp. 191–192.

SECTION 04

Term structure: spot, par & forward

Spot rates are zero-coupon yields — one rate per maturity — and they are the ideal term-structure dataset because there is nothing to reinvest

Where zero-coupon strips trade, spot rates are observed directly. Everywhere else, analysts bootstrap them out of coupon bonds.

Maturity	1y	2y	3y	4y	5y	7y	10y	20y	30y
Canada spot	0.31	0.57	0.80	0.96	1.11	1.30	1.58	1.98	2.06
Australia spot	0.03	0.07	0.30	0.59	0.81	1.17	1.52	2.35	2.47
Slope	↗	↗	↗	↗	↗	↗	↗	↗	↗

UPWARD-SLOPING

Long rates above short rates. A positive maturity premium and no expectation of falling short rates. The usual shape — both curves above.

FLAT

Every maturity at the same rate. Nothing is being expected — and the par and forward curves collapse onto the spot curve.

INVERTED

Long rates below short rates. The market is pricing cuts — historically a recession signal, always a financing-window signal.

¹ CFA Example 1: a five-year Canadian 1.00% bond priced on these spots is 99.50, and its YTM is 1.104% — just under the 1.11% five-year spot.

Source: CFA Institute, Curriculum Vol 6, LM 9, §2 "Maturity Structure of Interest Rates and Spot Rates", Exhibit 3 and Example 1, pp. 209–215.

A bond's no-arbitrage price is each cash flow discounted at its OWN spot rate — not all of them at a single YTM

A one-year cash flow and a ten-year cash flow are different assets. Pricing them at the same rate is a convenience, not a valuation.

$$PV = PMT / (1 + Z_1)^1 + PMT / (1 + Z_2)^2 + \dots + (PMT + FV) / (1 + Z_n)^n$$

where: Z_t = the spot rate for maturity t . For an issuer with credit risk, add a spread to every spot — which is precisely the Z-spread from §2.

WHY NO-ARBITRAGE

A dealer can strip the coupons and sell each as a separate zero. If the whole bond is worth less than the sum of its stripped parts, you buy the bond, strip it, and bank a riskless profit.

YTM IS STILL USEFUL

It compresses the whole curve into one quotable number, which is what makes bonds comparable and spreads computable. Just remember a curve is hiding behind it.

Source: CFA Institute, Curriculum Vol 6, LM 9, §2 "Bond Pricing Using Spot Rates", Equation 1, pp. 212–213.

Worked — a three-year 5% bond priced against spots of 2% / 3% / 4% is 102.960, and the single YTM that reproduces that price is 3.935%

The YTM is not an average of 2, 3 and 4. It is the PV-weighted blend — dominated by year three, where 91% of the present value sits.

THREE-YEAR 5% ANNUAL-COUPON BOND · PAR 100 · ANNUAL COMPOUNDING

Spot curve: $Z_1 = 2\%$, $Z_2 = 3\%$, $Z_3 = 4\%$. Cash flows 5, 5, 105. Price it on the spots, then back out the equivalent YTM.

1. Spot pricing: $5/(1.02)^1 = 4.902$ · $5/(1.03)^2 = 4.713$ · $105/(1.04)^3 = 93.345$ → Price = 102.960
2. The bond is at a premium: the 5% coupon beats every spot rate on the curve
3. Now solve one rate: $102.960 = 5/(1+r)^1 + 5/(1+r)^2 + 105/(1+r)^3$ → $r = 3.935\%$
4. Re-price at that single YTM: $4.811 + 4.629 + 93.520 = 102.960$ — the same total
5. But look at the pieces: year 1 is 4.902 on spots vs 4.811 on YTM; year 3 is 93.345 vs 93.520

Price 102.960 either way · YTM 3.935% · same total, different cash-flow PVs

A par rate is the coupon that makes a bond price at exactly 100 against the spot curve — and it is what the published Treasury curve actually shows

Par rates are used for term-structure work because they neutralise the tax, coupon and trading distortions that come with bonds priced away from 100.

$$100 = \text{PMT} / (1 + z_1)^1 + \dots + (\text{PMT} + 100) / (1 + z_n)^n$$

→ solve for PMT

where: Set price = face = 100 and solve for the coupon. That PMT divided by 100 is the par rate — and it is also that hypothetical bond's YTM.

PAR RATE vs YTM

A YTM belongs to a specific bond with a specific coupon. A par rate belongs to the CURVE — it is the coupon a new bond would have to carry to price at 100 today. On-the-run yields are close to par rates but not equal to them.

CURVE-SHAPE RULES

Upward-sloping spot curve → par curve sits BELOW it.
Flat spot curve → par curve equals it.
Downward-sloping spot curve → par curve sits ABOVE it.
(CFA Exhibit 8, p. 227.)

Source: CFA Institute, Curriculum Vol 6, LM 9, §3 "Par Rates from Spot Rates", Equation 2, pp. 217–218.

Worked — spots of 5.263 / 5.616 / 6.359 / 7.008% give par rates of 5.263 / 5.606 / 6.306 / 6.899%, each at or below its own spot

The one-year par rate is trivially the one-year spot. Beyond that, factor PMT out of the PV equation and divide — no iteration needed.

PAR RATES FROM A GIVEN SPOT CURVE · EFFECTIVE ANNUAL RATES

Spot curve: $Z_1 = 5.263\%$, $Z_2 = 5.616\%$, $Z_3 = 6.359\%$, $Z_4 = 7.008\%$. Find the coupon PMT that prices each bond at exactly 100.

1. 1-year: $100 = (\text{PMT} + 100)/1.05263 \rightarrow \text{PMT} = 5.263$. The one-year par rate is Z_1 .
2. Two-year discount factors: $1/1.05263 = 0.95000 \cdot 1/(1.05616)^2 = 0.89648$
3. 2-year: $\text{PMT} \times (0.95000 + 0.89648) = 100 - 89.648 = 10.352 \rightarrow \text{PMT} = 10.352 / 1.84648 = 5.606$
4. 3-year: add $1/(1.06359)^3 = 0.83115 \rightarrow \text{PMT} \times 2.67763 = 100 - 83.115 = 16.885 \rightarrow \text{PMT} = 6.306$
5. Verify: $5.606/1.05606 + 105.606/(1.05606)^2 = 100$ ✓ the par bond really does price at par

Par rates 5.263% · 5.606% · 6.306% · 6.899% — each at or below its own spot rate

An implied forward rate is the breakeven reinvestment rate that makes rolling a shorter zero exactly as good as holding a longer one

Naming: in "AyBy", A is how many years forward the period starts and B is the tenor of the rate. So 3y1y is the one-year rate, three years from today.

$$(1 + Z_A)^A \times (1 + \text{IFR}_{A,B-A})^{(B-A)} = (1 + Z_B)^B$$

where: Z_A = the shorter spot, Z_B = the longer spot, IFR = the implied $(B - A)$ -year rate starting at year A. Money markets state the same thing in months.

NOT A FORECAST

Expect the future rate ABOVE the forward → go short and roll. Expect it BELOW → go long and lock. The forward is the point where you do not care.

SPOT → FORWARD

Divide the longer compounded spot by the shorter one, then take the $(B - A)$ -th root. Every forward is a ratio of two adjacent points on the spot curve.

FORWARD → SPOT

Run it the other way: $(1 + Z_n)^n = (1 + 0y1y)(1 + 1y1y) \dots (1 + (n-1)y1y)$. The spot rate is the geometric average of the one-year forwards.

Source: CFA Institute, Curriculum Vol 6, LM 9, §3 "Forward Rates from Spot Rates", Equation 3 and Exhibits 4–5, pp. 218–220.

Worked — with a three-year zero at 3.65% and a four-year zero at 4.18%, the 3y1y implied forward rate is 5.79%

Question being answered: what extra return does the fourth year have to deliver to justify going out one year further? That number is the 3y1y.

COMPUTE THE 3Y1Y IMPLIED FORWARD RATE · ANNUAL COMPOUNDING

Three-year zero yields $Z_3 = 3.65\%$; four-year zero yields $Z_4 = 4.18\%$. Solve Equation 3 for $IFR_{3,1}$.

1. Set up: $(1 + Z_3)^3 \times (1 + IFR_{3,1})^1 = (1 + Z_4)^4$
2. Compound each side: $(1.0365)^3 = 1.113545$ and $(1.0418)^4 = 1.177979$
3. $1 + IFR_{3,1} = 1.177979 / 1.113545 = 1.057860$
4. $IFR_{3,1} = 5.786\% \approx 5.79\%$ — well above the 4.18% four-year spot
5. Sense check: the forward MUST exceed Z_4 whenever the spot curve is rising, or the average could not climb

3y1y = 5.79% — the year-4 rate at which rolling the 3-year zero ties buying the 4-year

HOW TO TRADE IT

Think the one-year rate three years out will be ABOVE 5.79%? Buy the three-year zero and roll — you beat the four-year. Think BELOW? Buy the four-year and lock it in. The forward is the indifference point, not a prediction.

Source: CFA Institute, Curriculum Vol 6, LM 9, §3 "Forward Rates from Spot Rates", Exhibit 4 illustration, pp. 218–220.

Spot, par and forward curves carry identical information — three lenses on one curve, and their ordering depends only on the slope

Know any one of the three and the other two are determined. There is no extra information in the forward curve — only a different view of the same data.

Spot curve shape	Forward curve	Par curve	What it is saying
Upward-sloping (normal)	Above spot	Below spot	Higher future short rates
Flat	Equal	Equal	Rates expected unchanged
Downward-sloping (inverted)	Below spot	Above spot	Lower future rates / cuts

WHY FORWARDS OVERSHOOT WHEN THE CURVE RISES

A long spot is the geometric average of the shorter spot and the forwards in between. If Z_3 exceeds Z_2 , the 2y1y forward must exceed Z_3 to drag that average up. Averages always lag the marginal input.

THE IDENTITY, WITH NUMBERS

CFA: forwards of 1.88 / 2.77 / 3.54 / 4.12% compound to spots of 1.880 / 2.324 / 2.728 / 3.074% (pp. 220–221).

Source: CFA Institute, Curriculum Vol 6, LM 9, §4 "Spot, Par, and Forward Yield Curves", Exhibits 6–8, pp. 225–227.

Worked — bootstrap a spot curve out of par bonds at 2%, 2.5% and 3%: $Z_1 = 2.000\%$, $Z_2 = 2.506\%$, $Z_3 = 3.020\%$

Bootstrapping is sequential substitution: solve Z_1 , plug it into the two-year equation to get Z_2 , plug both into the three-year equation to get Z_3 .

THREE PAR BONDS AT 100, ANNUAL COUPONS — RECOVER THE SPOTS

1-year 2.0% par bond · 2-year 2.5% par bond · 3-year 3.0% par bond. Each trades at exactly 100. Find Z_1 , Z_2 , Z_3 .

1. Z_1 : $100 = 102 / (1 + Z_1) \rightarrow 1 + Z_1 = 1.02 \rightarrow Z_1 = 2.000\%$ (a one-period bond IS a zero)
2. Z_2 : $100 = 2.5/1.02 + 102.5/(1 + Z_2)^2 \rightarrow 100 - 2.451 = 97.549 = 102.5/(1 + Z_2)^2 \rightarrow Z_2 = 2.506\%$
3. Check: Z_2 exceeds both Z_1 and the 2.5% par coupon — the curve is upward-sloping
4. Z_3 : $100 = 3/1.02 + 3/(1.02506)^2 + 103/(1 + Z_3)^3 \rightarrow 100 - 2.941 - 2.855 = 94.204 = 103/(1 + Z_3)^3$
5. Solve: $(1 + Z_3)^3 = 103 / 94.204 = 1.09337 \rightarrow Z_3 = 3.020\%$

Spot curve Z_1 2.000% · Z_2 2.506% · Z_3 3.020% — every spot above its own par rate ✓

Source: Worked exercise built on CFA Institute, Curriculum Vol 6, LM 9, §3, Equation 2, pp. 217–218 — the curriculum runs spot $\rightarrow$ par; here we run par $\rightarrow$ spot.

Deck 14 — Fixed Income II: Pricing, Yields & Term Structure

Price = PV of promised cash flows at the market rate.
Coupon below the rate → discount, equal → par, above → premium; the gap is the PV of the coupon shortfall.

Between coupon dates, full price = flat price + accrued interest. Dealers quote flat, and accrued interest depends on the calendar, never on the yield.

Yield-to-worst is the lowest of the YTM and every yield-to-call — the VIVU callable notes work out at 5.15%, the yield to the first call.

Money-market rates come as discount or add-on rates — convert both to a 365-day bond-equivalent yield. An FRN sits at par when required margin = quoted margin.

You earn the quoted YTM only if you hold to maturity, the issuer pays in full and on time, and you reinvest every coupon AT the YTM.

Matrix pricing, four steps: identify comparables, back out and average their yields, interpolate to the target tenor, discount the target's cash flows.

G-spread differences two yields; Z-spread adds a constant to every spot; OAS strips out the option — below Z for a callable, above Z for a putable.

Spot, par and forward curves are one information set. Upward-sloping curve: forwards above spots, spots above par rates. Bootstrap sequentially.